

COMMUNICATION SYSTEMS II

INTRODUCTION TO DIGITAL COMMUNICATION SYSTEMS

6 Hours/ week

2 Hrs. Theory, 2 Hrs. Tutorial, and 2 Hrs. Lab.

Textbooks

- 1. Modern Digital and Analog Communication Systems, 4th ed., by B.P Lathi & Zhi Ding, 2010.**
- 2. Digital Communications: Fundamentals and Applications, 3rd ed., by B. Sklar, & F. Harris, 2021.**

SYLLABUS

CHAPTER 6:

SAMPLING AND ANALOG-TO-DIGITAL CONVERSION

- SAMPLING THEOREM
- PULSE CODE MODULATION (PCM)
- DIGITAL MULTIPLEXING
- DIFFERENTIAL PULSE CODE MODULATION (DPCM)
- ADAPTIVE DIFFERENTIAL PCM (ADPCM)
- DELTA MODULATION

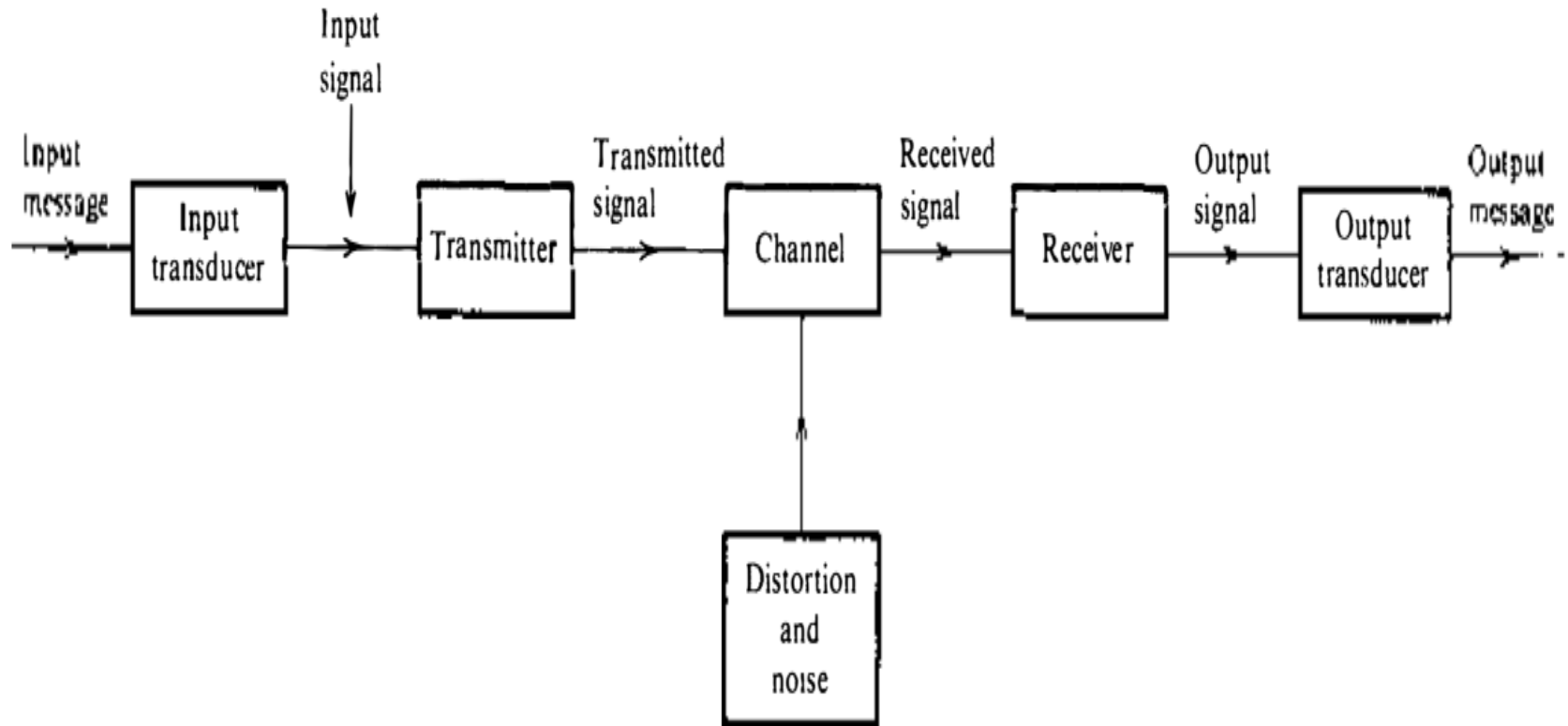
SYLLABUS

CHAPTER 7:

PRINCIPLE OF DIGITAL DATA TRANSMISSION

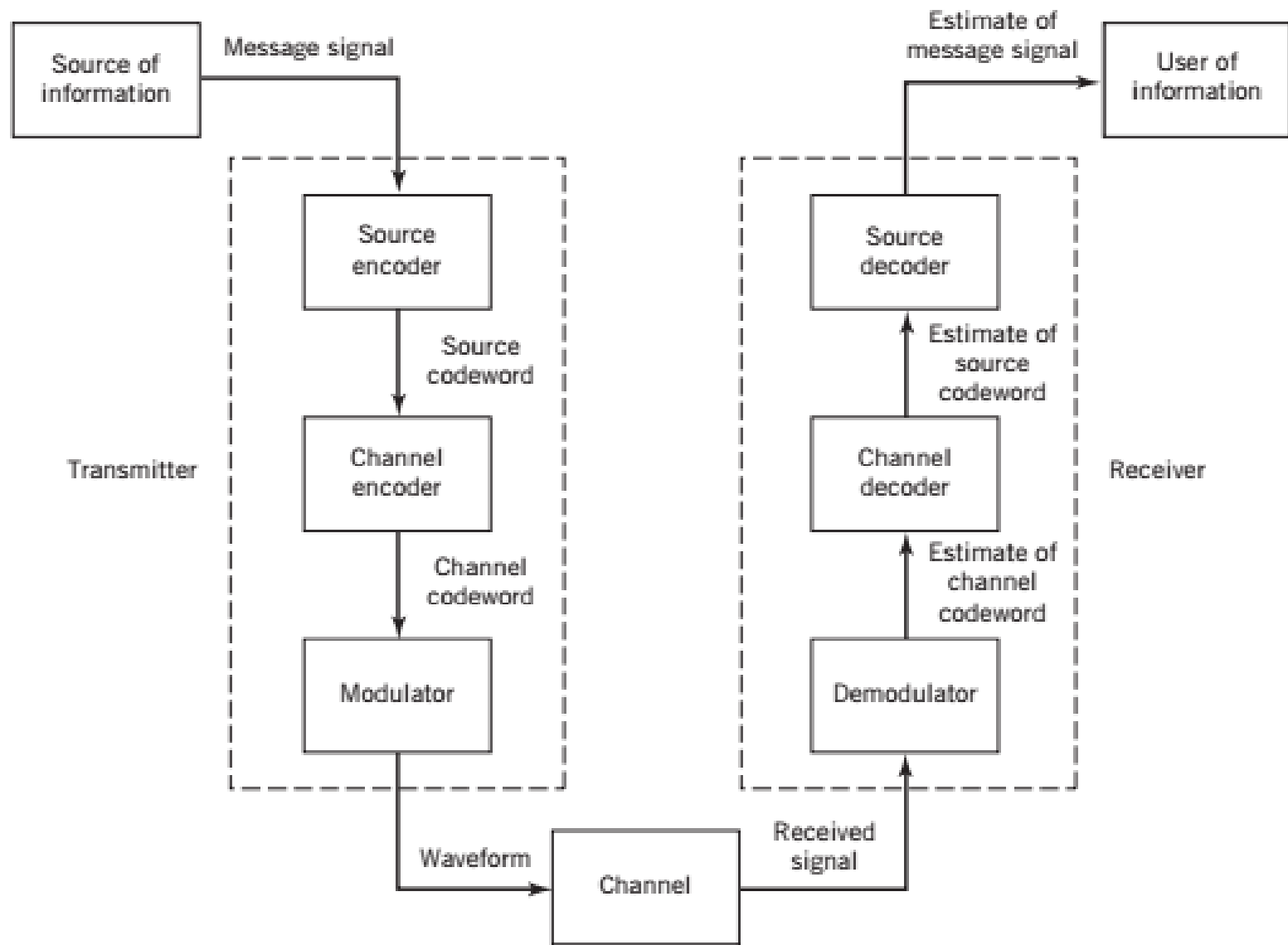
- DIGITAL COMMUNICATION SYSTEMS
- LINE CODING
- DIGITAL RECEIVERS AND REGENERATIVE REPEATERS
- PAM: M-ARY BASEBAND SIGNALING
- DIGITAL CARRIER SYSTEMS
- M-ARY DIGITAL CARRIER MODULATION

- A typical communication system is shown in Figure



- The source originates a message, such as a human voice, a television picture, an e-mail message, or data. If the data is nonelectric (e.g., human voice, e-mail text, a scene), it must be converted by an input transducer into an electric waveform referred to as the message signal through physical devices such as a microphone, a computer keyboard, or a camera, etc.
- The transmitter transforms the input (message) signal into an appropriate form for efficient transmission. The transmitter may consist of one or more subsystems: an analog-to-digital (A/D) converter, an encoder, and a modulator. Similarly, the receiver may consist of a demodulator, a decoder, and a digital-to-analog (D/A) converter.

- The channel is a medium of choice that can convey the electric signals at the transmitter output over a distance. A typical channel can be a pair of twisted copper wires (e.g., in telephone and DSL), coaxial cable (e.g. in television and Internet), an optical fiber, or a radio cellular link. Additionally, a channel can also be a point-to-point connection in a mesh of interconnected channels that form a communication network.
- The receiver reprocesses the signal received from the channel by reversing the signal transformation made at the transmitter and removing the distortions caused by the channel. The receiver output is passed to the output transducer, which converts the electric signal to its original form—the message.
- The destination is the unit where the message transmission terminates.



Digital Communication System

Advantages of Digital Communications

1. Digital signal can withstand channel noise and distortion much better than analog signal.
2. The viability of regenerative repeaters.
3. Digital hardware implementation is flexible and permits the use of microprocessors, digital switching, and large-scale integrated circuits.
4. Digital signals can be coded to yield extremely low error rates and high fidelity as well as for privacy.
5. It is easier and more efficient to multiplex several digital signals.
6. Digital communication is inherently more efficient than analog in exchanging SNR for bandwidth.
7. The cost of digital hardware continues to halve every two or three years, while performance or capacity doubles over the same time period.

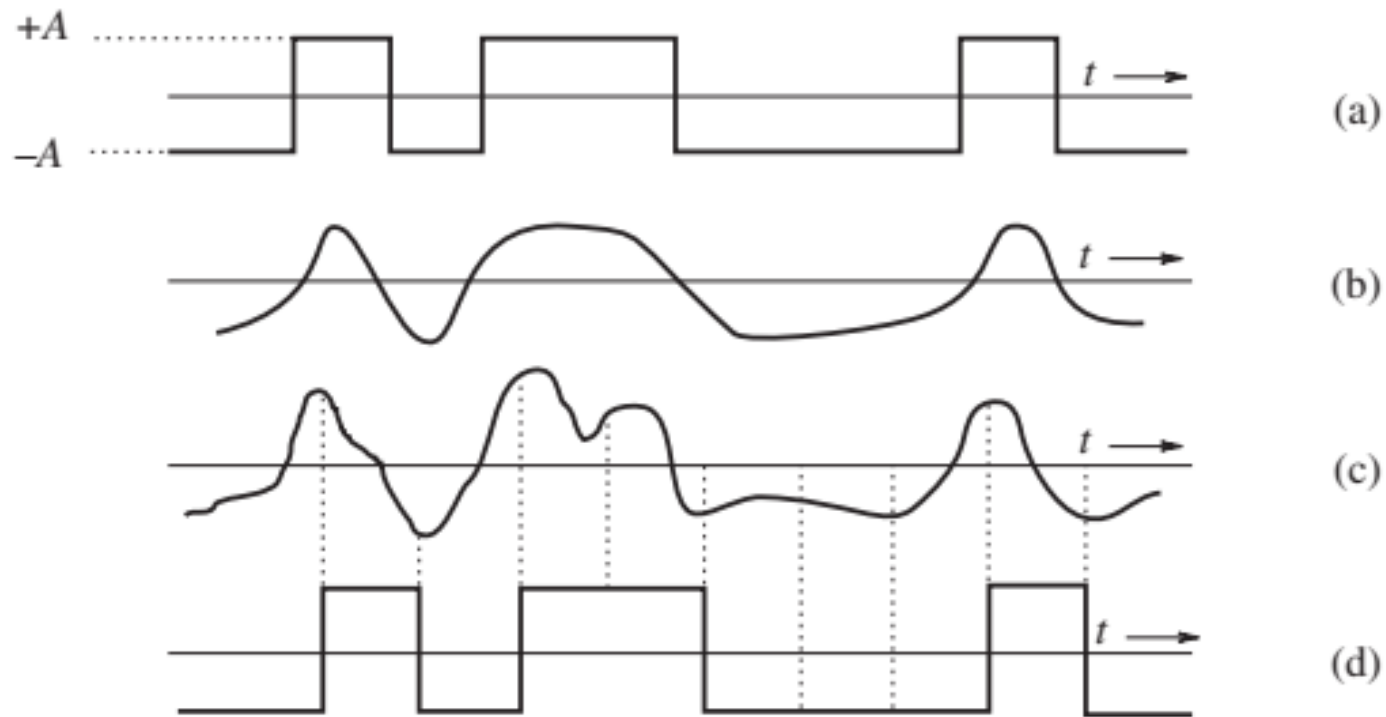


Figure 1.3

- (a) Transmitted signal.
- (b) Received distorted signal (without noise).
- (c) Received distorted signal (with noise).
- (d) Regenerated signal (delayed).